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The 'AI divide' between the Global North and the Global South

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The AI divide means the economic and social benefits of AI remain geographically concentrated, primarily in the Global North.
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- The economic and social benefits of AI remain geographically concentrated, primarily in the Global North.
 - Without an enabling operating environment, disparities in AI readiness will feed into global inequality.
 - 3 steps can help set the AI journey on the right path, including investment in fostering lasting benefits and promoting local education.
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Technological innovation is opening a new frontier in the development agenda for countries around the world. In recent years, artificial intelligence (AI) has had a profound impact on global issues in agriculture, healthcare, education and more. There is tremendous potential in harnessing the power of AI tools to increase economic growth, but the economic and social benefits of this technology remain geographically concentrated, primarily in the Global North.

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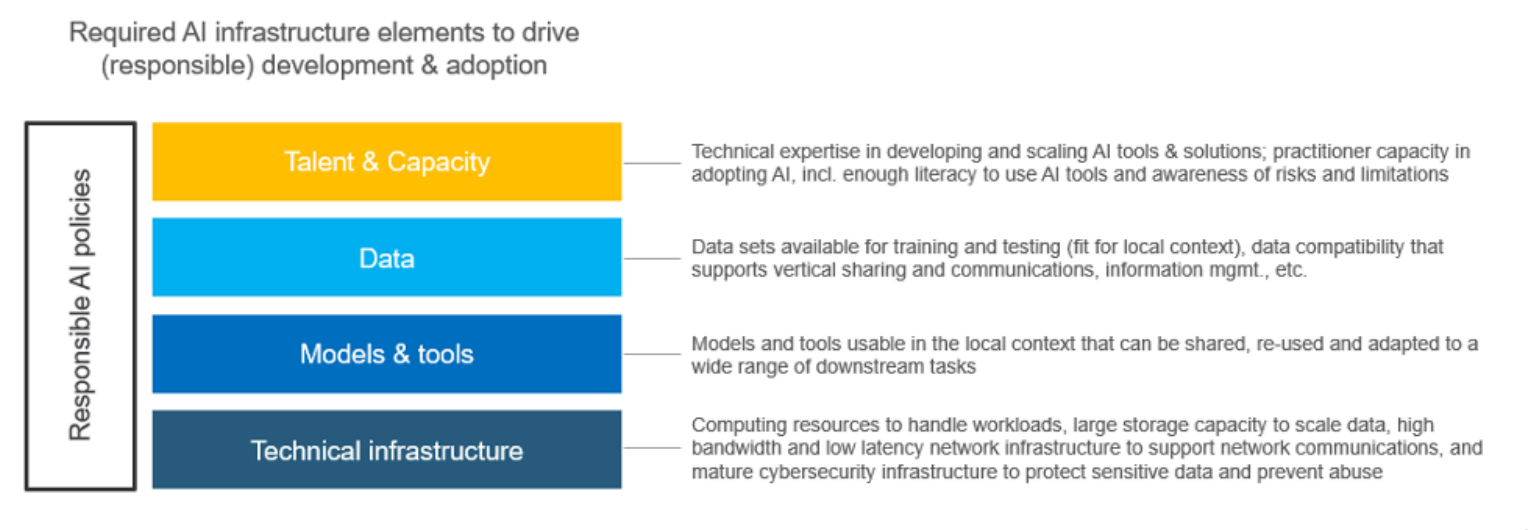
AI could contribute up to [\\$15.7 trillion to the global economy by 2030](#). While all regions of the global economy [stand to benefit from AI](#), North America and China will see the largest GDP gains. Countries in the Global South will experience more moderate increases due to the much lower rates of adoption of AI technologies. Unsurprisingly, developed nations with

use AI, as articulated in various national strategies such as [AIForAll from India](#).

Indeed, an [Oxford Insights assessment](#) of 181 countries around the world and their preparedness in using AI in public services highlights that the lowest-scoring regions include much of the Global South, such sub-Saharan Africa, some Central and South Asian countries, and some Latin American countries. The report emphasizes that governments require the appropriate operating environment in order to support AI development, which include a robust technology sector, adequate data infrastructure, and strategic vision and attention to governance and ethics at the state level. Without an enabling operating environment, disparities in AI readiness will feed into global inequality.

Structural limitations in the Global South

One of the root causes of this “AI divide” is found in structural limitations, as there are marked gaps between the Global North and Global South. Successful adoption of AI on a scaled level requires a demanding infrastructure that has elements of technical infrastructure, models and tools, data, and talent and capacity. On top of that, policies and guidelines are essential to ensuring trustworthiness in regulating new technologies.



In order to successfully implement and scale AI-driven solutions, sound technical infrastructure is necessary. This includes [high capacity computing resources](#) to handle the workloads, large storage capacity to scale as the data grows, a network infrastructure that is high bandwidth and low latency to support network communications, and a mature cybersecurity infrastructure to protect sensitive data and prevent abuse. The cost of implementing such advanced technologies is one of the biggest barriers for resource-

resource-constrained countries, let alone keeping it running and maintaining it long term.

Reusable models and tools are vital in accelerating AI research and development (R&D).

Foundation models that have been trained with a large set of unlabeled data and can be adapted to new downstream tasks are **fueling** the development of new capabilities and the deployment of new use cases. However, their R&D is generally dominated by a handful of companies in the US and China (due to large costs associated with the model training), and these companies limit who can access the trained models or to what extent these models can be used.

For specific AI solutions to be useful and responsible (i.e. trustworthy, inclusive and transparent), large amounts of local data is required for training and testing, to adapt to the local context and reflect unique social realities. Data availability and compatibility is a big pain point in resource-constrained regions, partly as a result of low infrastructure maturity and practitioner capacity. In the case of using real time data to make calculations based on foundation models, reliable network connectivity remains a major barrier.

Successful deployment of modern AI also depends on talent and capacity. Technical talent is needed to set up and maintain the technical infrastructure, explore and use existing models and tools, complete data sets, and develop new solutions and tools. Furthermore, industry practitioners also need to be upskilled to be able to adopt the new tools and solutions made possible by AI.

To make things more complex, AI comes with significant risks. Policies and guidelines are necessary to safeguard AI, ensuring its benefits and trustworthiness while mitigating its risks. Populations in the Global South are generally less literate on topics such as data privacy and algorithmic bias, and hence governance guidelines and policies need to be in place to guide the implementations of AI systems.

Three steps to be on the right path

When resource-constrained countries are unable to overcome the structural limitations to reap the rewards of AI opportunities, the gap between developed regions and resource-constrained regions will continue to widen. To decrease the gap, three steps can help set the AI journey on the right path.

algorithms to enable economic growth, the Global South is experiencing the rise of industries that engage low-skilled workers to perform data labelling and correction within the [AI value-chain](#).

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To help countries in the Global South break this pattern, local governments and international aid organizations need to reassess the local capabilities and plan investments strategically to uplevel the development trajectory of these countries and foster lasting benefits. For instance, investments can be made to improve basic AI infrastructure and capabilities, ensuring the prerequisites are in place to deploy highly beneficial solutions.

2. Promote local education and use open resources

Local education is key in creating a sustainable talent pipeline and bridging the capacity gap in AI. Massively Open Online Courses (MOOC), such as [Khan Academy](#), offer cost-effective, structured instruction for resource-constrained countries to uplift their AI education. Some offerings can be designed to work with offline-learning functionalities for low connectivity scenarios. Accordingly, infrastructure needs to have the basic capabilities to support experimentation and solution development in the learning process. Not only do academic and research institutions need computers with good network connections and power to run modern AI models, but they also need access to AI-ready datasets.

In tandem with investing in educational opportunities, special attention must be paid to the AI talent pool and pipeline. It is imperative to focus on skills development and build a sustainable, technical workforce. At the [Africa Development Center](#), for example, Microsoft is helping ensure all workers gain workplace skills to bolster the talent pipeline in Africa.

technology providers and development communities are essential to enabling a future-ready ecosystem. Governments should utilize the expertise and capacity of global technology providers as well local development communities to co-design the roadmap in meeting the key requirements for responsible deployment of AI. This includes identifying gaps and barriers in the local context, prioritizing, and defining a collaboration plan that enables the best synergy. This roadmap can be integrated into government AI strategies to steer countries in the right direction when establishing national AI ecosystems. The World Economic Forum's [National AI Strategy Peer Network](#) provides a good platform for policymakers and practitioners to share lessons, challenges and best practices.

AI applications have tremendous potential in [tackling economic and social challenges](#) around the world. However, the Global South is lagging behind due to structural limitations. Unless we act together to tackle the structural challenges, the “AI divide” will continue to widen between developed and resource-constrained countries, leaving much of the world behind.

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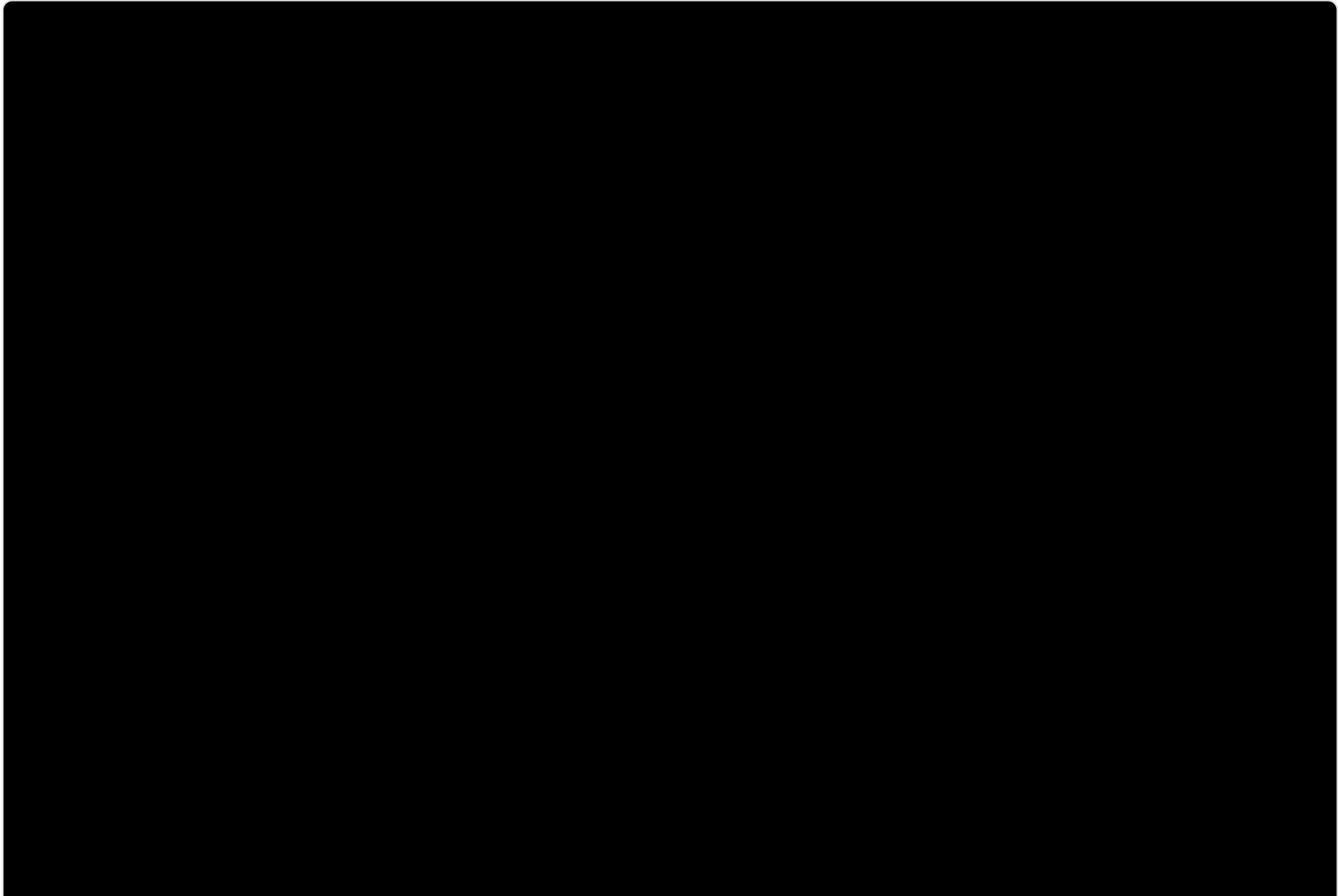
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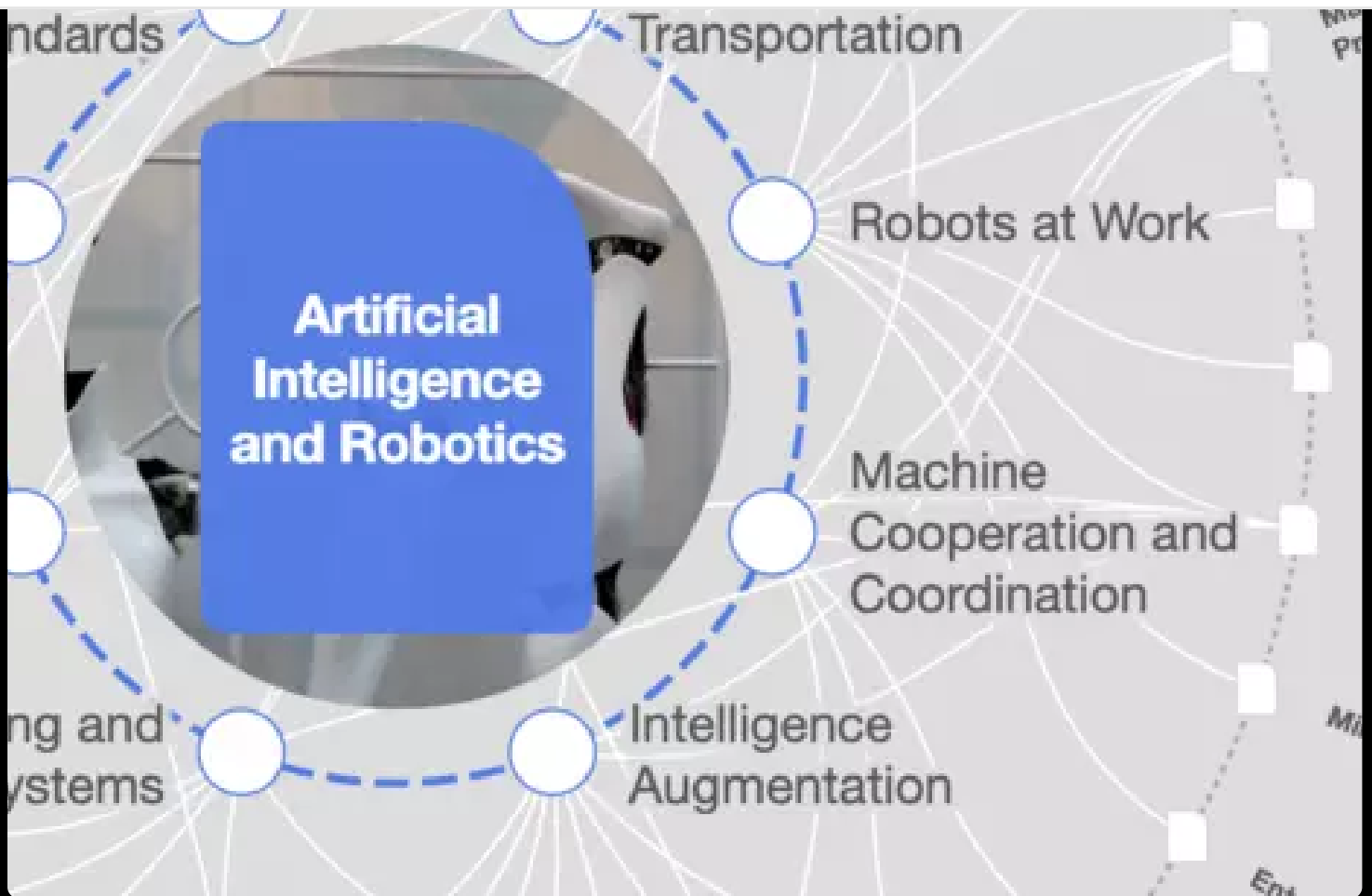
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